MiniOS Alpha — User Manual

Operating System: MiniOS Alpha | **Project:** IT 3131 Final Project | **Members:** PE, BALICAO, TUMAYAO, STO. THOMAS

1. Introduction

MiniOS Alpha is a simple command-line operating system simulator written in Python. It runs inside any terminal and simulates fundamental OS components, including:

- Command shell execution
- Hierarchical file management
- Process lifecycle tracking
- Dynamic memory allocation
- Role-based user management
- Graceful error handling

Note: MiniOS Alpha is an educational simulator designed to demonstrate operating systems concepts, not a real bootable operating system.

2. System Requirements

- **Operating System:** Windows (Command Prompt / CMD recommended), macOS, or Linux
- **Python Version:** Python 3.8 or higher
- **Dependencies:** None (uses standard Python library only: os, sys, time, datetime, json, hashlib, getpass, platform)

3. Installation

1. Download or clone the project repository:
   `git clone https://github.com/ancypng/minios-alpha.git`
   `cd minios-alpha`
2. Verify Python is installed and accessible:
   `python --version`
3. Confirm core project files exist in the directory:
   - main.py
   - filesystem.py
   - processes.py
   - memory.py

- users.py
- requirements.txt

4. How to Start MiniOS

1. Open Windows Command Prompt (CMD) or terminal and navigate to the project directory.
2. Launch the simulator:
   python main.py
3. MiniOS displays the boot banner and prompts for login:

```
==================================================
            MINIOS ALPHA v1.0
       Operating Systems Final Project
==================================================

Welcome to MiniOS!

Created by PE, BALICAO, MENDOZA, STP. THOMAS, TUMAYAO

==================================================
              MiniOS Login
==================================================
Demo accounts: admin / admin123, student / student123

Username: admin
Password:

[OK] Login successful. Welcome, admin!

Type 'help' to see the available commands.

Demo accounts: admin / admin123, student / student123
Username:
```

4. After logging in, the shell prompt appears:
   MiniOS Alpha>

5. Login Instructions

MiniOS Alpha requires authentication before granting shell access.

Default Accounts

Username	Password	Role
admin	admin123	Administrator
student	student123	Standard User

Login Steps

1. Enter admin or student at the Username: prompt and press Enter.
2. Enter the password at the Password: prompt. *(Note: Password characters are hidden for security.)*
3. On successful login, the welcome message is displayed:
   ```
   Login successful.
   Welcome, admin!
   Type 'help' to see the available commands.

   MiniOS Alpha>
   ```
4. **Attempt Limit:** The system allows up to 3 consecutive attempts. After 3 failed attempts, MiniOS shuts down automatically.
5. **Switching Users:** Use logout or login to return to the login screen.

6. Command List

MiniOS Alpha provides 24 commands and aliases divided into functional categories.

6.1 General Commands

COMMAND: help

- **Purpose:** Displays the list of all available commands organized by category.
- **Syntax:** help

```
MiniOS Alpha> help
Available Commands:
[General]
  help              Show this list
  clear             Clear the screen
  date / time       Show the date / the time
  whoami            Show the current user
  systeminfo        Show system information
```

```
  history             Show commands typed this session
  echo <text>         Print text
  version             Show the MiniOS version
  exit                Shut down MiniOS
[Files]
  dir [path]          List files and folders
  mkdir <name>        Create a directory
  create <file>       Create an empty file
  write <file> <text>  Write text into a file
  read <file>         Show a file's contents
  delete <name>       Delete a file or empty directory
  rename <old> <new>   Rename a file or directory
  cd <path>           Change directory (cd .. goes up)
  pwd                 Show the current directory
[Processes]
  run <name>          Start a simulated process
  process             List running processes
  kill <pid>          Terminate a process
[Memory]
  memory              Show memory status
  allocate <MB>       Allocate memory
  free <MB>           Release memory
[Users]
  users               List user accounts
  logout              Log out and return to login
  login               Switch user (alias of logout)
```

COMMAND: clear

- **Purpose:** Clears the terminal screen (cls is supported as an alias).
- **Syntax:** clear

```
MiniOS Alpha> clear
```

COMMAND: date

- **Purpose:** Displays the current system date.
- **Syntax:** date

```
MiniOS Alpha> date
Current Date: Thursday, October 22, 2026
```

COMMAND: time

- **Purpose:** Displays the current system time in 12-hour AM/PM format.
- **Syntax:** time

```
MiniOS Alpha> time
Current Time: 09:45:10 PM
```

COMMAND: whoami

- **Purpose:** Displays the username of the current logged-in user.
- **Syntax:** whoami

```
MiniOS Alpha> whoami
Current User: admin
```

COMMAND: systeminfo

- **Purpose:** Displays simulated hardware, operating system, and session information.

- **Syntax:** systeminfo

```
MiniOS Alpha> systeminfo
========================================
        SYSTEM INFORMATION
========================================
OS Name:      MiniOS Alpha
Version:      1.0
Architecture: 64-bit
CPU:          Virtual CPU
Memory:       1024 MB (256 MB used)
Host:         Windows (Python 3.11.4)
User:         admin (Administrator)
Processes:    0 running
Files/Dirs:   0 file(s), 2 folder(s)
Uptime:       1m 45s
========================================
```

COMMAND: history

- **Purpose:** Shows the history of commands entered during the current session.
- **Syntax:** history

```
MiniOS Alpha> history
  1  whoami
  2  dir
  3  systeminfo
  4  history
```

COMMAND: echo

- **Purpose:** Prints the specified text back to the screen.
- **Syntax:** echo <text>

```
MiniOS Alpha> echo Hello MiniOS
Hello MiniOS
```

COMMAND: version

- **Purpose:** Displays the MiniOS version.
- **Syntax:** version

```
MiniOS Alpha> version
MiniOS Alpha v1.0
IT 3131 Final Project
```

COMMAND: exit

- **Purpose:** Shuts down the MiniOS simulator (shutdown is supported as an alias).
- **Syntax:** exit

```
MiniOS Alpha> exit
Shutting down MiniOS...
Goodbye!
```

6.2 File Management Commands

COMMAND: pwd

- **Purpose:** Shows the current working directory path.
- **Syntax:** pwd

```
MiniOS Alpha> pwd
```

COMMAND: dir

- **Purpose:** Lists files and directories in the current folder or specified path (ls is supported as an alias).
- **Syntax:** dir [path]

```
MiniOS Alpha> dir
Directory of \
    <DIR>  Documents
       12  notes.txt
```

COMMAND: mkdir

- **Purpose:** Creates a new directory.
- **Syntax:** mkdir <folder>

```
MiniOS Alpha> mkdir Documents
Directory created.
```

COMMAND: create

- **Purpose:** Creates a new empty file.
- **Syntax:** create <file>

```
MiniOS Alpha> create notes.txt
File created.
```

COMMAND: write

- **Purpose:** Writes text into a file. Creates the file if it does not exist.
- **Syntax:** write <file> <text>

```
MiniOS Alpha> write notes.txt Operating Systems Final Project
File saved.
```

COMMAND: read

- **Purpose:** Displays the contents of a file.
- **Syntax:** read <file>

```
MiniOS Alpha> read notes.txt
Operating Systems Final Project
```

COMMAND: delete

- **Purpose:** Deletes a file or an empty directory.
- **Syntax:** delete <name>

```
MiniOS Alpha> delete notes.txt
```

```
File deleted.
```

COMMAND: rename

- **Purpose:** Renames an existing file or directory.
- **Syntax:** rename <old> <new>

```
MiniOS Alpha> rename notes.txt draft.txt
Renamed.
```

COMMAND: cd

- **Purpose:** Changes the current directory (cd .. goes to parent directory; cd \ returns to root).
- **Syntax:** cd <path>

```
MiniOS Alpha> cd Documents
Current directory: \Documents
MiniOS Alpha> cd ..
Current directory: \
```

6.3 Process Management Commands

COMMAND: run

- **Purpose:** Starts a new simulated process and assigns it a PID.
- **Syntax:** run <name>

```
MiniOS Alpha> run calculator
Process started.
PID: 1
Name: calculator
Status: RUNNING
```

COMMAND: process

- **Purpose:** Lists all currently running simulated processes.
- **Syntax:** process

```
MiniOS Alpha> process
PID    NAME          STATUS    STARTED
1      calculator    RUNNING   21:45:00
```

COMMAND: kill

- **Purpose:** Terminates a simulated process using its PID.
- **Syntax:** kill <pid>

```
MiniOS Alpha> kill 1
Process 1 terminated.
```

6.4 Memory Management Commands

COMMAND: memory

- **Purpose:** Displays current memory status including total, used, and free memory.

- **Syntax:** memory

```
MiniOS Alpha> memory
Memory Status----------------------
Total Memory: 1024 MB
Used Memory:   256 MB
Free Memory:   768 MB
```

COMMAND: allocate

- **Purpose:** Allocates a specified amount of memory in megabytes (MB).
- **Syntax:** allocate <MB>

```
MiniOS Alpha> allocate 200
200 MB allocated.
```

COMMAND: free

- **Purpose:** Releases a specified amount of user-allocated memory in megabytes (MB).
- **Syntax:** free <MB>

```
MiniOS Alpha> free 200
200 MB released.
```

6.5 User Management Commands

COMMAND: users

- **Purpose:** Lists all registered user accounts, their roles, and identifies the active user.
- **Syntax:** users

```
MiniOS Alpha> users
USER        ROLE
admin       Administrator  <- you
student     Standard User
```

COMMAND: logout

- **Purpose:** Logs out the current user and returns to the login screen (login is an alias).
- **Syntax:** logout

```
MiniOS Alpha> logout
Logging out admin...
========================================
            MiniOS Login
========================================
Demo accounts: admin / admin123, student / student123
Username:
```

7. Common Errors

MiniOS Alpha handles invalid inputs gracefully and does not crash when an error occurs. Below are the most common error messages, their causes, and resolutions:

#	Error Scenario	Trigger Input	Output Message	Resolution
1	Unknown Command	hellooo	Unknown command. Type 'help' to see the available commands.	Verify spelling or type help to view valid commands.
2	File Does Not Exist	read missing.txt	Error: File does not exist.	Use dir to verify the file exists in the current directory.
3	Entity Already Exists	mkdir Documents	Error: A file or directory with that name already exists.	Choose a different name or delete the existing entity.
4	Directory Not Empty	delete Documents	Error: Directory is not empty.	Delete the files inside the directory before deleting the directory.
5	Invalid Name	create report:final	Error: Invalid name. Avoid / \ : * ? " < > |	Remove forbidden special characters from the name.
6	Process Does Not Exist	kill 999	Error: Process does not exist.	Use process to find valid running PIDs.
7	Process Limit Reached	run app	Error: Process limit reached (10). Kill one first.	Use kill <pid> to terminate an existing process before starting a new one.
8	Insufficient Free Memory	allocate 1000	Error: Not enough free memory. Only 768 MB available.	Check memory and request an amount less than or equal to free memory.

#	Error Scenario	Trigger Input	Output Message	Resolution
9	Cannot Free Unallocated Memory	free 50	Error: Cannot free 50 MB. Only 0 MB is allocated by you.	You can only free memory that you previously allocated.
10	Missing Command Arguments	write notes.txt	Error: Missing argument. Usage: write <file> <text>	Supply all required parameters indicated by the usage hint.

8. How to Exit MiniOS

Method 1: The exit Command

Type exit or shutdown at the prompt:

```
MiniOS Alpha> exit
Shutting down MiniOS...
Goodbye!
```

Method 2: Keyboard Interrupts

Press **Ctrl+C**, **Ctrl+D**, or **Ctrl+Z** (on Windows). The system intercepts the interrupt and exits gracefully:

```
Shutting down MiniOS...
Goodbye!
```

State Persistence

- **Filesystem:** Files and folders persist across reboots via minios_disk.json.
- **Processes and Memory:** Simulated processes and dynamic memory allocations are volatile and reset cleanly to default upon restart.